

Open-access single-cell and spatial omics datasets for ARDS and IPF

Publicly available human single-cell and spatial transcriptomics datasets for acute lung injury (ARDS/ALI) and idiopathic pulmonary fibrosis (IPF) are concentrated in a handful of landmark studies, almost all published between 2019 and 2025. For ARDS, the field is dominated by COVID-19 lung studies—true non-COVID ARDS single-cell data from human lung tissue remains essentially absent from public repositories. For IPF, four foundational scRNA-seq datasets form the backbone of the IPF Cell Atlas (~551,000 cells), ([American Physiological So...](#)) while a burst of spatial transcriptomics studies in 2024–2025 has added critical spatial context. Every dataset below has been verified as open-access (processed data freely downloadable), with GEO accessions confirmed as public. Controlled-access-only datasets (e.g., those solely on EGA or dbGaP without an open tier) are excluded but noted for completeness at the end.

Acute lung injury datasets span COVID-19 autopsy, BALF, and spatial profiling

The ARDS/ALI single-cell landscape is overwhelmingly COVID-19-derived. The most impactful studies profiled either **lung autopsy tissue** (Melms, Delorey, Bharat) or **bronchoalveolar lavage fluid** (Liao, Grant, Ren). Spatial profiling of ARDS lungs uses primarily NanoString GeoMx DSP and 10x Visium platforms. **No published open-access scATAC-seq dataset from human ARDS lung tissue exists as of early 2026.**

#	Study	Year	Journal	DOI	Data Type	Platform	Ref
1	Melms et al.	2021	<i>Nature</i> 595:114–119	10.1038/s41586-021-03569-1	snRNA-seq	10x Chromium	C5
2	Delorey et al.	2021	<i>Nature</i> 595:107–113	10.1038/s41586-021-03570-8	snRNA-seq + spatial (GeoMx DSP)	10x Chromium; NanoString GeoMx WTA/CTA	C6
3	Liao et al.	2020	<i>Nature Medicine</i> 26:842–844	10.1038/s41591-020-0901-9	scRNA-seq + scTCR-seq	10x Chromium 5'	C7
4	Grant et al.	2021	<i>Nature</i> 590:635–641	10.1038/s41586-020-03148-w	scRNA-seq	10x Chromium	C8
5	Ren et al.	2021	<i>Cell</i> 184:1895–1913	10.1016/j.cell.2021.01.053	scRNA-seq	10x Chromium 5' and 3'	C9

#	Study	Year	Journal	DOI	Data Type	Platform	Ref
6	Bharat et al.	2020	<i>Sci. Transl. Med.</i> 12:eabe4282	10.1126/scitranslmed.abe4282	scRNA-seq	10x Chromium	C
7	Cross et al.	2023	<i>JCI Insight</i> 8:e157837	10.1172/jci.insight.157837	Spatial transcriptomics	NanoString GeoMx DSP (CTA + COVID spike-in)	C
8	Lee, Barnett et al.	2025	<i>Nature Communications</i> 16:1979	10.1038/s41467-025-56473-x	snRNA-seq + spatial (GeoMx WTA + Visium)	10x Chromium; NanoString GeoMx; 10x Visium FFPE	5
9	Margaroli et al.	2021	<i>Cell Reports Medicine</i> 2:100242	10.1016/j.xcrm.2021.100242	Spatial transcriptomics	NanoString GeoMx DSP (1,860 genes)	M
10	Mothes et al.	2023	<i>Nature Communications</i> 14:791	10.1038/s41467-023-36333-2	Spatial transcriptomics + imaging	10x Visium; multiplex IF; LSFM	Z

#	Study	Year	Journal	DOI	Data Type	Platform	
11	Rendeiro et al.	2021	Nature 593:564–569	10.1038/s41586-021-03475-6	Spatial proteomics (IMC, 36 proteins)	Hyperion Imaging Mass Cytometry	

Notable controlled-access ARDS datasets (excluded per criteria but important to acknowledge): Wendisch et al. 2021 (*Cell* 184:6243–6261; [PubMed](#) EGAS00001005634) profiled COVID-ARDS BAL and compared to IPF. Chua et al. 2020 (*Nature Biotechnology* 38:970–979; EGAS00001004481) characterized airway epithelium-immune interactions. [Nature](#) Wauters et al. 2021 (*Cell Research* 31:272–290; EGAS00001004717) deep-profiled BAL from 31 COVID patients. [Nature](#) All three are on EGA with controlled access only.

Critical gap: non-COVID ARDS. No open-access scRNA-seq or snRNA-seq dataset from human lung tissue of non-COVID ARDS (sepsis-ARDS, trauma-ARDS, aspiration-ARDS) was identified. Grant et al. 2021 provides the closest approximation with its non-COVID pneumonia comparison cohort, [Nature](#) but this is flow cytometry and bulk RNA-seq, not single-cell. [Nature](#) The NU SCRIPT study has additional data on dbGaP (phs002300.v1.p1) but under controlled access. [PubMed Central](#)

IPF datasets form a mature ecosystem from foundational atlases to cutting-edge spatial maps

IPF single-cell genomics is built on **four landmark scRNA-seq datasets** (Adams, Habermann, Reyfman, Morse) that together form the IPF Cell Atlas. [American Physiological So...](#) Between 2024 and 2025, three major spatial transcriptomics studies (Franzén, Mayr, Vannan) added critical tissue-context information. **One snATAC-seq study (Valenzi et al. 2023) provides the only open chromatin profiling of human IPF lung tissue.** [PubMed Central](#)

#	Study	Year	Journal	DOI	Data Type	Platform
1	Adams et al.	2020	<i>Science Advances</i> 6:eaba1983	10.1126/sciadv.aba1983	scRNA-seq	10x Chromium
2	Habermann et al.	2020	<i>Science Advances</i> 6:eaba1972	10.1126/sciadv.aba1972	scRNA-seq	10x Chromium
3	Reyfman et al.	2019	<i>AJRCCM</i> 199:1517–1536	10.1164/rccm.201712-2410OC	scRNA-seq	10x Chromium
4	Morse et al.	2019	<i>Eur. Respir. J.</i> 54:1802441	10.1183/13993003.02441-2018	scRNA-seq	10x Chromium V2
5	Xu et al.	2016	<i>JCI Insight</i> 1:e90558	10.1172/jci.insight.90558	scRNA-seq (epithelial)	Fluidigm C1 (Smart-seq)

#	Study	Year	Journal	DOI	Data Type	Platform
6	Tsukui et al.	2020	<i>Nature Communications</i> 11:1920	10.1038/s41467-020-15647-5	scRNA-seq (mesenchymal)	10x Chromium
7	Carraro et al.	2020	<i>AJRCCM</i> 202:1540–1550	10.1164/rccm.201904-0792OC	scRNA-seq (epithelial)	10x Chromium
8	Winters et al.	2022	<i>JCI Insight</i> (PMC)	—	scRNA-seq	10x Chromium
9	Valenzi et al.	2023	<i>Eur. Respir. J.</i>	—	snATAC-seq + scRNA-seq	10x Chromium (snATAC + scRNA)
10	Franzén et al.	2024	<i>Nature Genetics</i> 56:1725–1736	10.1038/s41588-024-01819-2	Spatial (Visium)	10x Visium (fresh frozen)
11	Mayr et al.	2024	<i>Science Advances</i> 10:ead15473	10.1126/sciadv.adl5473	Spatial (Visium FFPE + Xenium)	Visium CytAssist; 10x Xenium; COMET IF

#	Study	Year	Journal	DOI	Data Type	Platform
12	Vannan et al.	2025	<i>Nature Genetics</i> 57:647–658	10.1038/s41588–025–02080–x	Spatial (Xenium + Visium HD)	10x Xenium (343–gene panel); 10x Visium HD
13	Watanabe, Yoshida et al.	2025	<i>Eur. Respir. J.</i>	—	scRNA–seq + Spatial (Visium)	10x Chromium; 10x Visium
14	Blumhagen, Kurche et al.	2023	<i>Respiratory Research</i> 24:287	10.1186/s12931–023–02572–6	Spatial (GeoMx DSP)	NanoString GeoMx DSP (whole transcriptome)

Integrated atlases bridge ARDS and IPF under a unified lung reference

Two meta-resources provide essential cross-disease context and should be considered foundational for any ARDS or IPF analysis.

The Human Lung Cell Atlas (HLCA), published by Sikkema et al. in *Nature Medicine* (2023; DOI: 10.1038/s41591–023–02327–2), integrates **49 scRNA-seq datasets encompassing ~2.4 million cells from 486 individuals** across 15+ diseases including COVID-19, pulmonary fibrosis, COPD, and healthy lung. (Nature) The core atlas (~584,900 cells from 107 healthy individuals) provides consensus cell-type annotations (Humancellatlas) for 61 cell types. (Helmholtz Munich) (GitHub) A key finding is that **SPP1+ profibrotic monocyte-derived macrophages share transcriptional signatures across COVID-19, IPF, and lung carcinoma**, (Helmholtz Munich) unifying these diseases

mechanistically. (Nature) The data is freely available on CellxGene (humancellatlas) (Humancellatlas) (collection 6f6d381a–7701–4781–935c–db10d30de293) under CC BY 4.0.

The IPF Cell Atlas (Neumark et al. 2020, *Am J Physiol Lung Cell Mol Physiol* 319:L887–L892; DOI: 10.1152/ajplung.00451.2020) integrates the four foundational IPF scRNA–seq datasets (Yale School of Medicine) (Adams GSE136831, Habermann GSE135893, (Frontiers) Reyfman GSE122960, Morse GSE128033) into a unified interactive portal at ipfcellatlas.com.

(Yale School of Medicine) (American Physiological So...) It comprises ~**551,000 cells from 134 individuals** (American Physiological So...) (52 IPF, 18 COPD, 54 controls, plus other ILD subtypes).

(American Physiological So...) The Kaminski Lab at Yale also maintains related atlases including the COVID Cell Atlas (covidcellatlas.com) and the Lung Endothelial Cell Atlas (lungendothelialcellatlas.com). (Yale School of Medicine) (American Physiological So...)

Additional resources include **LungMAP** (lungmap.net), which hosts CellRef v1.0 (Guo et al. 2023, *Nature Communications* 14:4566; 48 human lung cell types) (Nature) and provides healthy lung references with snRNA–seq and snATAC–seq (American Journal of Respir...) (used for COVID–19 susceptibility gene analysis), (eLife) and the **Travaglini et al. 2020** healthy lung atlas (*Nature* 587:619–625; Synapse syn21041850; [CellxGene](https://cellxgene.org)), which defined 58 cell populations as the foundation for disease comparisons. (Nature)

Key technological and biological gaps remain significant

Several patterns emerge from this compilation that are important for researchers planning new studies or secondary analyses.

Non–COVID ARDS is a critical blind spot. Every open–access single–cell dataset from ARDS lung tissue or BALF is derived from COVID–19 patients. No published, publicly available scRNA–seq or snRNA–seq dataset exists from human lung tissue in sepsis–ARDS, trauma–ARDS, aspiration–ARDS, or other non–COVID etiologies. Grant et al. 2021 partially addresses this with bulk transcriptomics from 211 non–COVID pneumonia patients, (Nature) but single–cell resolution for non–COVID ARDS remains unavailable. (Nature) This limits our ability to distinguish COVID–specific pathology from general ARDS biology.

scATAC–seq coverage is minimal. Only one human IPF lung snATAC–seq dataset exists (Valenzi et al. 2023, 3 IPF + 2 controls). (PubMed Central) No scATAC–seq or multiome dataset from human ARDS lung tissue has been published. Wang et al. (*eLife* 2021; LungMAP) profiled snATAC–seq on ~90,980 (eLife) nuclei from non–diseased lungs, providing a reference, (eLife) but disease–state chromatin accessibility remains largely uncharted territory.

Spatial transcriptomics is maturing rapidly for IPF but not ARDS. Three Nature Genetics/Science Advances papers in 2024–2025 (Franzén, Mayr, Vannan) provide comprehensive spatial maps of IPF with increasing resolution (Visium → Visium HD → Xenium subcellular). For ARDS, spatial data is more fragmented, relying primarily on NanoString GeoMx DSP studies with smaller cohorts. **No published MERFISH or seqFISH study of human ARDS or**

IPF lung tissue was identified. 10x Xenium is emerging as the dominant imaging-based platform for lung disease. (MDPI +3)

Sample source matters for interpretation. Nearly all IPF scRNA-seq data comes from **end-stage transplant explants**, representing UIP pattern at its most advanced. (Science) Only Reyfman et al. 2019 included one cryobiopsy from earlier-stage (Omicardi) disease. (American Physiological So... (PubMed Central) For ARDS, data comes from either **autopsy tissue** (late/lethal disease) or **BALF** (which captures immune cells but not structural lung architecture). (Nature) Early-stage disease sampling remains underrepresented in both conditions.

Conclusion

This compilation identifies **11 open-access ARDS/ALI datasets and 14 open-access IPF datasets** meeting the strict criteria of human-only, publicly downloadable single-cell or spatial omics data, plus 2 major integrated atlases spanning both diseases. The field is anchored by a small number of highly impactful studies: for ARDS, the Melms, (Nature) Delorey, and Liao datasets collectively provide **>250,000 cells** from COVID-19 lungs; for IPF, the Adams and Habermann datasets alone contribute **>425,000 cells**. The most transformative recent additions are the 2024–2025 spatial transcriptomics studies (Franzén, Mayr, Vannan) that localize IPF cell states to specific tissue niches with subcellular resolution, and the Lee et al. 2025 integration of spatial profiling across DAD stages in COVID-19 lungs. The convergence of the HLCA's discovery that SPP1+ macrophages are shared across COVID-19 and IPF with the spatial localization of these cells in fibrotic niches represents the current frontier—and underscores why cross-disease analysis using these open datasets is now both feasible and scientifically compelling.